

MTU_ValueService **Technical Documentation**

BlueLine Waterjet
for series 2000
Common Rail
ECS-5, MCS-5

Application Marine

Functional Description
Operating Instructions
Workshop Manual
Installation and Commissioning Instructions

E532228/00E



Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

Diese Veröffentlichung einschließlich aller ihrer Teile ist urheberrechtlich geschützt. Jede Verwertung oder Nutzung bedarf der vorherigen schriftlichen Zustimmung der MTU Friedrichshafen GmbH. Das gilt insbesondere für Vervielfältigung, Verbreitung, Bearbeitung, Übersetzung, Mikroverfilmungen und die Einspeicherung und / oder Verarbeitung in elektronischen Systemen, einschließlich Datenbanken und Online-Diensten. Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Änderungen bleiben vorbehalten.

Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Imprimé en Allemagne

© 2006 Copyright MTU Friedrichshafen GmbH

Tout droit réservé pour cet ouvrage dans son intégralité. Toute utilisation ou exploitation requiert au préalable l'accord écrit de MTU Friedrichshafen GmbH. Ceci s'applique notamment à la reproduction, la diffusion, la modification, la traduction, l'archivage sur microfiches, la mémorisation et / ou le traitement sur des systèmes électroniques, y compris les bases de données et les services en ligne.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

© 2006 Copyright MTU Friedrichshafen GmbH

Esta publicación se encuentra protegida, en toda su extensión, por los derechos de autor. Cualquier utilización de la misma, así como su reproducción, difusión, transformación, traducción, microfilmación, grabación y/o procesamiento en sistemas electrónicos, entre los que se incluyen bancos de datos y servicios en línea, precisa de la autorización previa de MTU Friedrichshafen GmbH.

El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

Nos reservamos el derecho de introducir modificaciones.

Stampato in Germania

© 2006 Copyright MTU Friedrichshafen GmbH

Questa pubblicazione è protetta dal diritto d'autore in tutte le sue parti. Ciascun impiego o utilizzo, con particolare riguardo alla riproduzione, alla diffusione, alla modifica, alla traduzione, all'archiviazione in microfilm e alla memorizzazione o all'elaborazione in sistemi elettronici, comprese banche dati e servizi on line, deve essere espressamente autorizzato per iscritto dalla MTU Friedrichshafen GmbH.

Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

Impresso na Alemanha

© 2006 Copyright MTU Friedrichshafen GmbH

A presente publicação, inclusive todas as suas partes, está protegida pelo direito autoral. Qualquer aproveitamento ou uso exige a autorização prévia e por escrito da MTU Friedrichshafen GmbH. Isto diz respeito em particular à reprodução, divulgação, tratamento, tradução, microfilmagem, e a memorização e/ou processamento em sistemas eletrônicos, inclusive bancos de dados e serviços on-line.

Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

Wichtig – Important – Importante

Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen GmbH zurücksenden.

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

Veillez séparer la carte “Signalisation de mise en service” et la renvoyer à la MTU Friedrichshafen GmbH.

Les informations contenues dans la signalisation de mise en service constituent la base pour l'assistance en exploitation contractuelle (garantie, rechanges, etc.).

Rogamos separen la tarjeta “Aviso de puesta en servicio” y la devuelvan rellena a MTU Friedrichshafen GmbH.

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

Ritagliare “Avviso di messa in servizio” e rispedirlo debitamente compilato alla MTU Friedrichshafen GmbH.

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).

É gentileza cortar o cartão “Participação da colocação em serviço”, preenchê-lo e devolvê-lo a MTU Friedrichshafen.

Os dados referentes à colocação em serviço representam a base para o suporte logístico (garantia, peças sobressalentes, etc.) estabelecido contratualmente.



Postcard

MTU Friedrichshafen GmbH
Department SCSD
88040 Friedrichshafen
GERMANY

Bitte in Blockschrift ausfüllen!
Please use block capitals!
Prière de remplir en lettres capitales!
¡A rellenar en letras de imprenta!
Scrivere in stampatello!
Favor preencher com letras de forma!



Motornr.: Engine No.: N° du moteur: N° de motor: Motore N.: No. do motor:	Auftragsnr.: MTU works order No.: N° de commande: N° de pedido: N. commessa: No. do pedido:	Inbetriebnahme- meldung Commissioning Note
Motortyp: Engine model: Type du moteur: Tipo de motor: Motore tipo: Tipo do motor:	Inbetriebnahmedatum: Date put into operation: Mise en service le: Fecha de puesta en servicio: Messa in servizio il: Data da colocação em serviço:	Notice de mise en service Aviso de puesta en servicio
Eingebaut in: Installation site: Lieu de montage: Lugar de montaje: Installato: Incorporado em:	Schiffstyp / Schiffshersteller: Vessel/type/class / Shipyard: Type du bateau / Constructeur: Tipo de buque / Constructor: Tipo di barca / Costruttore Tipo de embarcação/estaleiro naval:	Avviso di messa in servizio Participação da colocação em serviço
Endabnehmer/Anschrift: End user's address: Adresse du client final: Dirección del cliente final: Indirizzo del cliente finale: Usuário final/endereço:		
Bemerkung: Remarks: Remarques: Observaciones: Commento: Observações:		

General Conditions	07
1 Safety	09
1.1 General conditions	09
1.2 Personnel and organizational requirements	10
1.3 Safety precautions when working on the engine	11
1.4 Auxiliary materials, fire prevention and environmental protection	14
1.5 Standards for warning notices in the publication	16
Functional Description	17
2 Product Summary	19
2.1 BlueLine system – Overview	19
2.2 BlueLine subsystems – Use	21
3 Subsystems	23
3.1 Remote Control System RCS-5	23
3.1.1 RCS-5 BlueLine – Overview	23
3.1.2 RCS-5 BlueLine – Use of the devices	25
3.1.3 RCS-5 BlueLine – Main control stand 1	29
3.1.4 RCS-5 BlueLine – Main control stand 2	32
3.1.5 RCS-5 BlueLine – Slave control stand	33
3.2 Monitoring and Control System MCS-5	35
3.2.1 MCS-5 BlueLine – Overview	35
3.2.2 MCS-5 BlueLine – Use of the devices	37
3.2.3 MCS-5 BlueLine – Main control stand	40
3.2.4 MCS-5 BlueLine – Slave control stand	44
3.3 Engine Control System ECS-5	45
3.3.1 ECS-5 BlueLine – Overview	45
3.3.2 ECS-5 BlueLine – Use of the devices	51
Operating Instructions	55
4 Monitoring and Control System MCS-5	57
4.1 Controls and Displays	57
4.1.1 MCS-5 – Controls and displays	57
4.1.2 Display – Overview of the screen pages	62
4.2 Operation	72
4.2.1 Switching on the overall BlueLine system	72
4.2.2 Adjusting brightness of indicators / instruments	73
4.2.3 Starting the engine	74
4.2.4 Override function – Activation	75
4.2.5 Override function – Deactivation	76
4.2.6 Stopping the engine	77
4.2.7 Emergency engine stop	78
4.2.8 BlueLine System – Switching off	79

4.2.9	Pump control – Control and display elements	80
4.2.10	Priming Pump Controller PPC – Function check	81
5	Remote Control System RCS-5	83
5.1	Controls and Indicators (standard version)	83
5.1.1	Command unit for one shaft	83
5.1.2	Command unit for two shafts	85
5.1.3	Command unit for three shafts	87
5.1.4	Command unit for four shafts	90
5.2	Controls and Indicators (special version)	92
5.2.1	Operating station and rotary encoder modules	92
5.3	Operation	95
5.3.1	Allocating initial command	95
5.3.2	Command transfer	96
5.3.3	Operation	97
5.3.4	Changing engine speed without engaging	98
5.3.5	Single Control Lever mode (SCL) – Activation	99
5.3.6	Single Control Lever mode (SCL) – Deactivation	100
5.3.7	Trolling mode – Activation	103
5.3.8	Trolling mode – Deactivation	104
6	Engine Control System ECS-5	105
6.1	Controls and Displays	105
6.1.1	LOP	105
6.1.2	Local Operating Station LOS and Local Operating Panel LOP – Controls and displays	106
6.2	Normal Operation: LOS or LOP with Display	108
6.2.1	Switching the engine ready for operation	108
6.2.2	Local mode – Activation	109
6.2.3	Starting the engine	110
6.2.4	Local mode – Engaging/disengaging the gearbox	111
6.2.5	Local mode – Changing engine speed	112
6.2.6	Stopping the engine	113
6.2.7	Local mode – Deactivation	114
6.2.8	Interlocking engine start	115
6.2.9	Emergency engine stop	116
6.3	Operation during Servicing: LOP without Display	117
6.3.1	Starting the engine	117
6.3.2	Stopping the engine	118
6.3.3	Interlocking engine start	119
6.3.4	Emergency engine stop	120
7	Troubleshooting	121
7.1	Faults in the BlueLine system	121
7.2	Troubleshooting	122
7.3	Alarms – General information	126
7.4	Alarm – Acknowledging at the control stand	127

7.5	Fault messages on the LOP display	128
7.6	Fault indication on printed circuit board MPU 29	133
7.7	ECU 4 fault codes	137
7.8	Oil priming pump, oil refill pump – Troubleshooting	146
8	Task Descriptions	149
8.1	Cabling, General for Engine / Gearbox / Unit	149
8.1.1	Engine wiring check	149
8.2	General Display and Controls	150
8.2.1	Indicator lamp - LED replacement	150
Workshop Manual		153
9	Task Descriptions	155
9.1	Operating Tests	155
9.1.1	BlueLine test mode – Execution	155
9.1.2	Overspeed test	158
9.1.3	Emergency stop test	159
9.1.4	Barring the engine	160
9.2	Checks and Settings	161
9.2.1	DIS settings – Check	161
9.2.2	Display DIS – Setting	162
9.2.3	PIM 4 node number – Check	163
9.2.4	Serial interface terminator plugs – Check	164
9.2.5	Jumper configuration of MCS-5 control unit – Check	165
9.2.6	CAN bus connections PIM 4 – Check	166
9.2.7	CAN bus line resistance – Check	168
9.2.8	MPU 29 node address – Check	170
9.2.9	Control stand configuration in Local Operating Panel LOP – Check	171
9.2.10	Shaft number setting – Check	173
9.2.11	LOP power supply – Check	174
9.2.12	Jumper configuration of LOP – Check	177
9.2.13	GCU 6 jumper configuration – Check	178
9.2.14	GCU 3 jumper configuration – Check	179
9.2.15	GCU node number – Check	180
9.2.16	MPU 29 node address – Setting	181
9.2.17	Configuration	182
9.2.18	MPU 29 node address in LOP – Setting	184
9.2.19	MPU 29 node address in LOP – Check	185
9.3	Repair Work	186
9.3.1	PIM 4 cover – Removal	186
9.3.2	PIM 4 cover – Installation	187
9.3.3	Command unit – Replacement	188
9.3.4	Rotary encoder module – Replacement	189
9.3.5	Operating panel – Replacement	190
9.3.6	Analog display instrument VDO OceanLine – Replacement	191
9.3.7	Lamp in analog display instrument VDO OceanLine – Replacement	192
9.3.8	Indicator lamp – Replacement	193

9.3.9	Indicator lamp – Removal	194
9.3.10	Indicator lamp – Installation	195
9.3.11	LED in indicator lamp – Replacement	196
9.3.12	Contactor – Replacement	198
9.3.13	LED in pushbutton/switch – Replacement	199
9.3.14	Display DIS – Replacement	201
9.3.15	PIM 4 parameters – Reset	203
9.3.16	Fuse in control unit PIM 4 – Replacement	204
9.3.17	Control unit PIM 4 – Replacement	205
9.3.18	Printed circuit board in a PIM cassette – Replacement	207
9.3.19	Fuse on printed circuit board in a PIM cassette – Replacement	208
9.3.20	Printed circuit board MPU 29 – Removal	210
9.3.21	Printed circuit board MPU 29 – Installation	211
9.3.22	Printed circuit board MPU 29 – Replacement	212
9.3.23	Data module on MPU 29 – Replacement	214
9.3.24	Data module on MPU 29 – Transfer	215
9.3.25	LOP fuse – Replacement	216
9.3.26	LOP with display front flap – Replacement	218
9.3.27	Local Operating Panel LOP – Replacement	220
9.3.28	Local Operating Station LOS – Replacement	221
9.3.29	Gear Control Unit GCU – Replacement	222
9.3.30	Priming Pump Controller PPC – Repair	224
9.3.31	Priming Pump Controller PPC – Replacement	226
9.4	Supplementary Technical Information	227
9.4.1	PIM 4 RCS – Internal design	227
9.4.2	Control unit PIM 4 MCS – Internal design	232
9.4.3	Local Operating Panel LOP – Design	237
9.4.4	Gear Control Unit GCU – Design	247
9.4.5	Priming Pump Controller PPC – Design	252
Installation and Commissioning Instructions		253
10	Installation	255
10.1	Preparation	255
10.1.1	Cable routes and openings between all installation locations – Check	255
10.1.2	Cables between installation locations – Routing	256
10.2	Mechanical Installation	258
10.2.1	PIM 4 module housing – Installation	258
10.2.2	OceanLine analog display instruments – Installation	259
10.2.3	VDO horn – Installation	261
10.2.4	Indicator lamp – Installation	262
10.2.5	Illuminated pushbutton / key switch – Installation	263
10.2.6	GCU – Installation	265
10.2.7	Priming Pump Controller PPC – Installation	266
10.2.8	DIS – Installation	267
10.2.9	LOP – Installation	268
10.2.10	LOS – Installation	270
10.2.11	Installing command unit ROS 11	272
10.2.12	Installing command unit ROS 7 / ROS 9	274

10.2.13	Installing rotary encoder modules ROS 10/P and ROS 10/S	277
10.2.14	ROS 10/T Remote Operating Station – Installation	278
10.2.15	Terminal box X010 – Installation	280
10.3	Electric Installation	281
10.3.1	HSK cable bushing – Preparation	281
10.3.2	Terminal strip – Preparation	282
10.3.3	PIM 4 cover – Removal	283
10.3.4	Connecting GCU 3	284
10.3.5	Connecting a third-party gearbox	286
10.3.6	Connecting engine governor ECU	289
10.3.7	Connecting display instruments on the main control stand	290
10.3.8	Main control stands – Connecting controls and status indicator lamps	291
10.3.9	Indicator lamps for single-point alarms – Connection	292
10.3.10	Connecting devices on a slave control stand	293
10.3.11	Connecting PIM 4 MCS control unit	294
10.3.12	Priming Pump Controller PPC (3-phase) – Connection	298
10.3.13	Starter and battery-charging generator – Connection	300
10.3.14	Barring tool limit switch – Connection	306
10.3.15	Main control stand display (option) – Connection	307
10.3.16	LOP – Connection	308
10.3.17	Connecting additional Yard signals (optional)	311
10.3.18	LOS – Connecting	313
10.3.19	Command units – Connecting on propulsion systems with three shafts	314
10.3.20	Command units – Connecting on propulsion systems with four shafts	317
10.3.21	Connecting command unit ROS 7 / ROS 9	319
10.3.22	Rotary encoder module and remote operating station – Connecting	320
10.3.23	Connecting PIM 4 RCS control unit	322
10.3.24	RCS extension module – Installation in PIM 4 RCS	325
10.3.25	Priming Pump Controller PPC (single-phase) – Connection	326
10.3.26	Terminal box X010 – Connection (battery-charging generator charges starter battery)	328
11	Initial Operation	331
11.1	Preparation	331
11.1.1	Checks prior to startup	331
11.1.2	Operating voltage – Initial application	332
11.1.3	Supply voltage distribution – Check	334
11.2	Settings	337
11.2.1	Configuration	337
11.2.2	Making settings via Mini Dialog	339
11.3	Tests	347
11.3.1	Initial engine startup at service interface on LOP	347
11.3.2	Initial engine startup at display on Local Operating Panel LOP or Local Operating station LOS	348
11.3.3	Initial engine startup at MCS	349
11.3.4	Testing speed control and clutch setting	350
11.3.5	Testing command transfer (only when several control stands are included)	351
11.3.6	System reset	352

Annex	353
12 Annex	355
12.1 Conversion tables	355
12.2 Abbreviations	361
12.3 Index	368